Natural Language Processing Course Project

Project Proposal | Track A: Survey

Forgotten or Merely Hidden? A Survey of Machine Unlearning for Large Language Models through the Lens of Forgetting Depth

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1. Team Members and Responsibilities

Member	Responsibilities
Yaqi Luo	Unlearning Methods & Taxonomy: Survey LLM machine unlearning methods, develop a technical taxonomy, and analyze the underlying principles of different approaches; write the corresponding sections of the proposal, midterm report, and final report; prepare and present the methods-related slides.
Jing Wang	Comparative Analysis & Challenges: Survey evaluation metrics and benchmarks, compare existing unlearning methods in terms of forgetting effectiveness, model utility, robustness, and applicability based on prior studies, and identify current limitations and challenges; write the corresponding sections of each report; prepare and present the related slides.
Shared	Project Planning & Integration: Define research questions, collect and review literature, refine the taxonomy, conduct critical analysis, and identify future research directions; jointly prepare, integrate, and revise the proposal, midterm deliverables, and final report; collaborate on the final slides, presentation, Q&A preparation, and submission.

2. Background and Motivation

Large language models (LLMs) are trained on massive datasets that may contain sensitive personal information, copyrighted content, and harmful knowledge. As concerns about privacy, copyright, and AI safety continue to grow, machine unlearning has emerged as an important research direction. Its goal is to remove the influence of specific training data or knowledge from a trained model without the computational cost of retraining it from scratch, while preserving the model's general capabilities.

However, existing unlearning methods differ significantly in their underlying mechanisms, effectiveness, and robustness against knowledge recovery. More importantly, a model's inability to produce certain information does not necessarily mean that the information has been truly forgotten, as supposedly removed knowledge may still be recovered through techniques such as adversarial prompting, fine-tuning, or other recovery attacks. This raises a fundamental question: *Does machine unlearning actually remove knowledge, or merely suppress its expression under limited evaluation settings?* Motivated by this distinction, our survey aims to systematically categorize existing LLM

unlearning methods, compare their strengths and limitations, and examine their effectiveness and robustness in achieving reliable forgetting through the lens of forgetting depth.

3. Problem Definition and Research Questions

LLM machine unlearning aims to remove the influence of specific training data or knowledge from a pretrained model while preserving its general capabilities. However, existing methods vary in how they achieve forgetting, and conventional evaluations may not fully reveal whether forgotten knowledge can be recovered under different conditions. This survey investigates LLM machine unlearning through the lens of forgetting depth and robustness, focusing on the following research questions:

1. **Methods and Taxonomy:** What are the main approaches to LLM machine unlearning, and how can they be systematically categorized based on their underlying mechanisms?
2. **Effectiveness and Robustness:** How do existing methods compare in terms of forgetting effectiveness, model utility, and robustness against knowledge recovery attacks that attempt to restore forgotten information?
3. **Challenges and Future Directions:** To what extent can current methods achieve genuine and robust forgetting rather than merely suppressing knowledge, and what challenges remain in evaluating and improving forgetting depth?

4. Expected Outcomes

We will write an 8-page survey on the robustness of LLM unlearning, using the ACL template. The survey will include:

1. A taxonomy of stress tests organized by input, representation and weight level.
2. An analysis of each level: which stress tests recover forgotten knowledge, under what conditions, against which unlearning methods, and why unlearning fails them.
3. A comparison across papers of how much forgotten knowledge each stress test recovers. For each entry we record the model, benchmark and metric, and flag results that are not directly comparable.
4. A review of defenses, grouped by the level of stress test they withstand, highlighting levels that no current defense addresses.
5. Open problems and future directions, including robustness in reasoning and multilingual models and evaluation protocols that cover all three levels.

4.1 Preliminary Taxonomy

An unlearned model may appear to have forgotten the target knowledge under standard evaluation, yet still reveal it under other conditions. Following OpenUnlearning, we call the procedures that test for this **stress tests**. They include deliberate attacks as well as routine operations such as quantization and fine-tuning. We classify stress tests by the level of the model they act on; each level requires more access than the one before it.

1. **Input level: query access only.** The model is left unchanged; only how it is queried or used changes.
   - Rephrased questions and in-context prompts
   - Adversarial prompts and jailbreaks
   - Queries in another language
   - Changes to decoding or reasoning mode, e.g. disabling the reasoning trace

Why unlearning fails here: it is trained on a fixed set of questions and answers, so it often does not generalize to new phrasings, other languages, or outputs other than the final answer.

2. **Representation level: access to internal activations, weights unchanged.**
   - Probing hidden states
   - Soft prompts optimized in the embedding space
   - Activation steering

   Why unlearning fails here: the model's internal representations still encode the knowledge; unlearning mainly stops it from being expressed in the output.

3. **Weight level: access to modify the weights.**
   - Quantization
   - Relearning on related data
   - Fine-tuning on unrelated data
   - Other weight tampering

   Why unlearning fails here: it changes the weights only slightly, so quantization or a small amount of further training can undo the change.

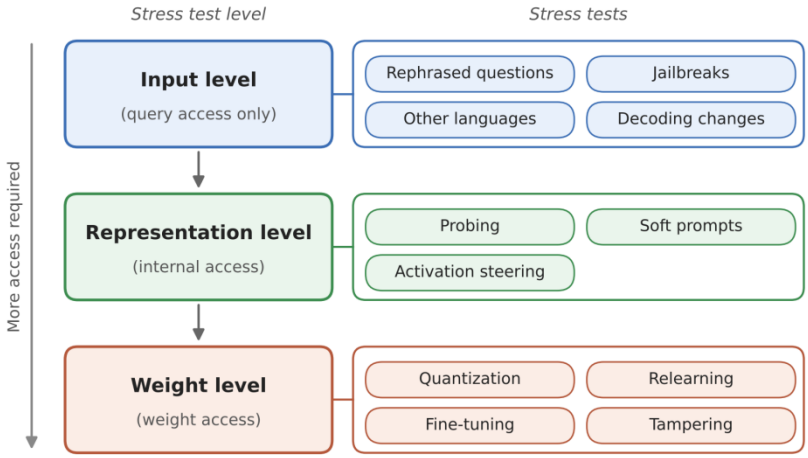


Figure 1: Taxonomy of stress tests for LLM unlearning, organized by the level of the model they act on. Lower levels require more access to the model.

Defenses are discussed after the taxonomy, grouped by the level of stress test they withstand.

4.2 Paper List

Input-level stress tests

- R-TOFU: Unlearning in Large Reasoning Models (EMNLP 2025)

Representation-level stress tests

- Soft Prompt Threats: Attacking Safety Alignment and Unlearning in Open-Source LLMs through the Embedding Space (NeurIPS 2024)

Weight-level stress tests

- Catastrophic Failure of LLM Unlearning via Quantization (ICLR 2025)
- Unlearning or Obfuscating? Jogging the Memory of Unlearned LLMs via Benign Relearning (ICLR 2025)

- Do Unlearning Methods Remove Information from Language Model Weights? (2024)

Multi-level evaluations

- Eight Methods to Evaluate Robust Unlearning in LLMs (2024) (input, representation)
- Does Unlearning Truly Unlearn? A Black Box Evaluation of LLM Unlearning Methods (2024) (input, weight)
- An Adversarial Perspective on Machine Unlearning for AI Safety (TMLR 2025) (input, representation, weight)
- Model Tampering Attacks Enable More Rigorous Evaluations of LLM Capabilities (TMLR 2025) (representation, weight)
- Multilingual Unlearning in LLMs: Transfer, Dynamics, and Reversibility (2026) (input, representation)
- LLM Unlearning Under the Microscope: A Full-Stack View on Methods and Metrics (2025) (input, weight)
- OpenUnlearning: Accelerating LLM Unlearning via Unified Benchmarking of Methods and Metrics (NeurIPS 2025) (representation, weight)

Defenses

- Towards LLM Unlearning Resilient to Relearning Attacks: A Sharpness-Aware Minimization Perspective and Beyond (ICML 2025)
- Latent Adversarial Training Improves Robustness to Persistent Harmful Behaviors in LLMs (TMLR 2025)
- Tamper-Resistant Safeguards for Open-Weight LLMs (ICLR 2025)
- Distillation Robustifies Unlearning (NeurIPS 2025)

Background: unlearning methods and benchmarks

- Machine Unlearning (IEEE S&P 2021)
- Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning (2024)
- TOFU: A Task of Fictitious Unlearning for LLMs (2024)
- The WMDP Benchmark: Measuring and Reducing Malicious Use with Unlearning (ICML 2024)

5. Project Timeline

1. **Oct 13–26 (2 weeks): Background reading.** Both members read the papers on unlearning methods, the benchmarks and the related surveys.
2. **Oct 27–Nov 16 (3 weeks): In-depth reading of stress-test papers.** For each paper we record the stress tests used, the unlearning methods tested, the model, the benchmark, the metric, and how much forgotten knowledge was recovered.
   - Yaqi Luo: input-level and representation-level papers, plus the multi-level evaluations that focus on these two levels.
   - Jing Wang: weight-level papers, plus the multi-level evaluations that focus on weight-level tests.
3. **Nov 17–23 (1 week): Revise the taxonomy.** Revise the three-level taxonomy based on the papers read, and add papers found in their reference lists.
4. **Midterm (expected late November):** Present the papers read, the revised taxonomy and a partial comparison across papers.

5. **Nov 24–30 (1 week): Defenses.** Read the six defense papers and record which levels of stress test each one withstands.
6. **Dec 1–21 (3 weeks): Analysis and writing.** Complete the comparison across papers, explain why unlearning fails at each level, list open problems, and write the full report in the ACL template.
7. **Dec 22–early January: Revision and presentation.** Revise the report, prepare the slides, and submit all materials.